

forecast price volatility and price covariance. What about the volume statistics—ADV and V_t ? How are these variables determined?

During the parameter estimation phase we showed how to measure these variables over the historical sample period to incorporate them into the non-linear regression model. What about our forecasting needs? Are the volume patterns expected to be exactly the same in the future as they were in the past? The answer is likely no. If historical variable values are used for the forward-looking forecast with differing market conditions then the actual costs could be much different than the forecasted estimates even with the correct model form and actual parameters.

For example, suppose there was a sudden spike in volatility and a decrease in liquidity, similar to what was experienced during the financial crisis and debt crisis. If the cost model did not incorporate these new input variables, a higher cost execution strategy would be selected as a result of dramatically underestimating costs. Furthermore, many times portfolio managers are interested in future expected costs under various market conditions. Managers typically buy shares under the most favorable conditions, such as low volatility and high liquidity, and sell shares under very dire circumstances, such as high volatility, low liquidity, and decreasing prices. It is actually these market circumstances at times that help managers decide whether to buy or sell shares. So even if the model is formulated correctly and the parameters are exact, incorporating the incorrect volume and volatility values will lead to inaccurate cost estimates. This chapter will provide the necessary techniques to forecast daily V_t and monthly ADV conditions.

TRADING STRATEGY

Algorithmic trading makes use of three types of trading strategies: percentage of volume POV , trading rate α , and trade schedule x_k

Let,

X = total shares to trade

V_t = expected volume during the trading horizon (excluding shares from the order)

The trading strategy variables are:

Percentage of Volume

$$POV = \frac{X}{X + V_t} \quad 0\% \leq POV \leq 100\% \quad (7.4)$$